

Question 2: Template Matching Implementation

Pushpendra Uikey

September 12, 2026

Overview

This section details a custom template matching algorithm utilizing Normalized Cross Correlation (NCC). The algorithm searches for a "No Parking" sign (`templateNoPark.png`) within a larger scene (`parking.png`). Because NCC is robust to global brightness shifts, it is highly effective at finding structural matches even if the lighting in the scene differs from the template.

Methodology

The algorithm was implemented in two distinct phases to handle background interference.

Phase 1: Unmasked NCC

In standard NCC, the entire template is compared to every possible patch in the image.

1. **Template Normalization:** The mean of the template is subtracted from every pixel to achieve zero-mean, and the template standard deviation (t_std) is calculated.
2. **Cross-Correlation:** The zero-mean template is cross-correlated with the image using a fast Fourier transform (`scipy.signal.correlate` with `mode='valid'`).
3. **Local Patch Normalization:** To ensure the correlation score is normalized to a $[-1, 1]$ range, the standard deviation of every local image patch is calculated dynamically. This is achieved efficiently by cross-correlating a matrix of ones with the image and the squared image.

Phase 2: Masked NCC

The standard template has a white background outside the red ring, while the sign in the actual scene has a blue background. This background mismatch drags down the NCC score. To fix this, a binary mask is generated to isolate the circular sign.

1. **Mask Generation:** Pixels in the template that are not pure white are flagged as 1.0 (the sign), while the white background is flagged as 0.0.
2. **Masked Statistics:** The mean and standard deviation of both the template and the local image patches are calculated *exclusively* using the pixels where the mask equals 1.0. The cross-correlation is computed between the image and the masked, zero-mean template.

Results

In all output visualizations, a red cross marks the pixel with the highest NCC score in that specific color channel.

Unmasked Evaluation

As specified, the scene was reduced by a factor of 5, and the template was scaled to three small sizes: 41×41 , 51×51 , and 61×61 . Because the template's white background conflicts with the scene's blue background, the peak matches (red crosses) fail to consistently localize the true sign across all channels. The peak wanders randomly across the image.

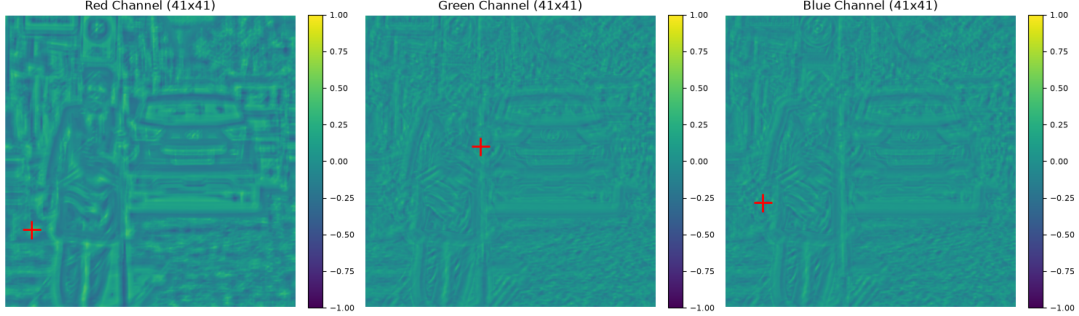


Figure 1: Unmasked NCC using a 41×41 template.

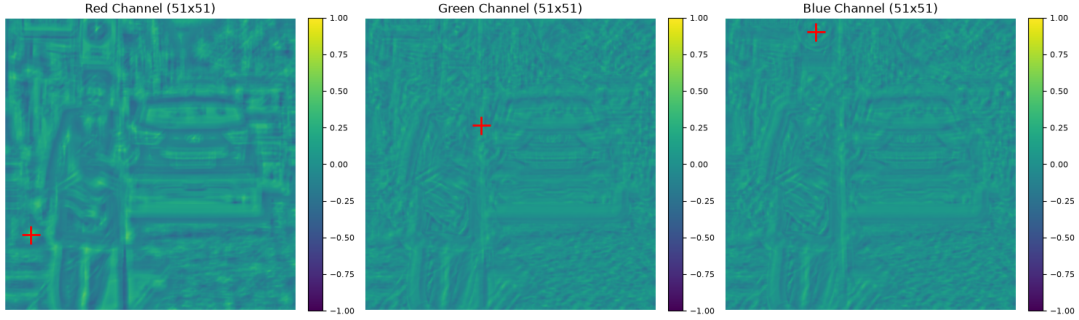


Figure 2: Unmasked NCC using a 51×51 template.

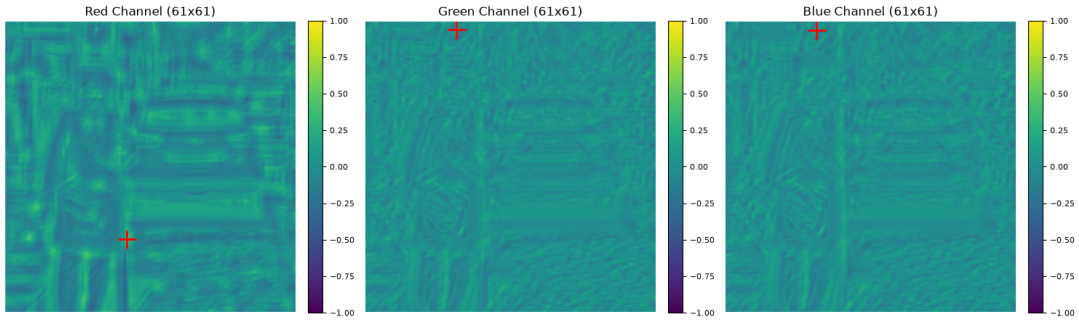


Figure 3: Unmasked NCC using a 61×61 template.

Masked Evaluation

The scene was left at its original resolution, and the template was scaled to three large sizes: 201×201 , 251×251 , and 301×301 . By mathematically ignoring the background corners of the template bounding box, the red channel perfectly and consistently localizes the "No Parking" sign in the top-middle of the image across two larger scales. Because the sign is predominantly red and white, the red channel provides the strongest structural match.

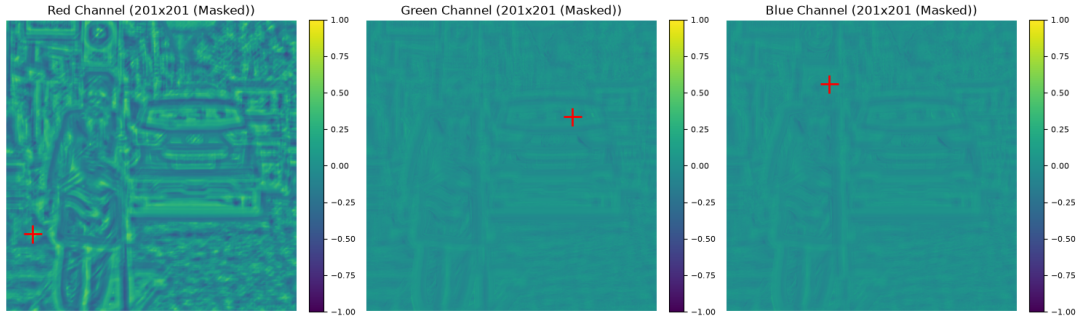


Figure 4: Masked NCC using a 201×201 template. The red channel correctly finds the sign.

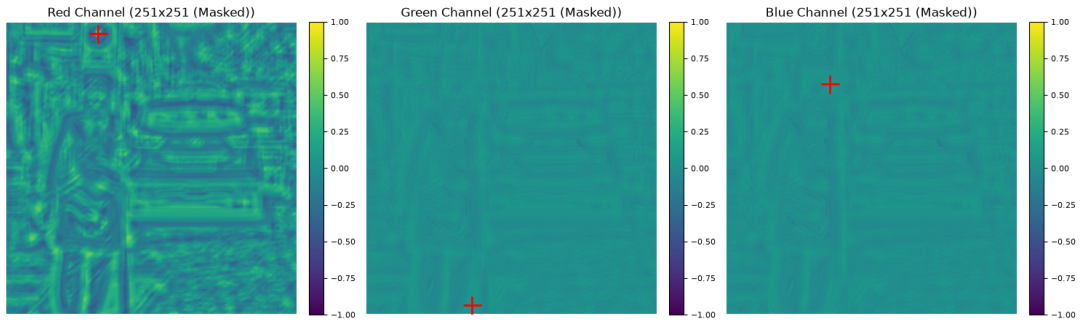


Figure 5: Masked NCC using a 251×251 template.

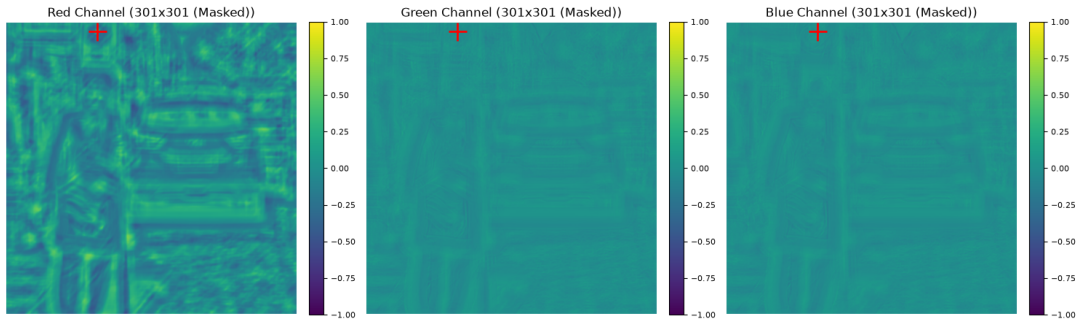


Figure 6: Masked NCC using a 301×301 template.